The Fokker-Planck Equation: The Deterministic Twin of a Stochastic Process

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Black-Scholes tells you the price of an option. It says nothing about where the underlying will be tomorrow.

For that, you need a completely different object, not a price, not a trajectory, but a probability density. A function that tells you, at every point in space and time, how probable it is that your process ends up there.

That density evolves according to the Fokker-Planck equation.

It is the deterministic twin of a stochastic process. While the SDE describes a single random path, Fokker-Planck describes the entire cloud of all possible paths at once, tracking how probability mass flows forward in time, pushed by drift and spread by diffusion.

Every model you use in quantitative finance: Black-Scholes, Heston, Vasicek, SABR, has a Fokker-Planck equation running underneath it. When you calibrate a local vol surface, you are implicitly inverting one. When you price a barrier option by PDE, you are solving one. When you run a Monte Carlo simulation, you are approximating its solution one path at a time.

Vol.6 of the Journal of a (Mini) Quant covers the equation, its derivation from Itô's lemma, the duality with Feynman-Kac, the stationary distribution, and where it actually lives inside the models you work with every day.

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Carlo Acerbi · 2nd
ADIA | Risknowledge SA | LARIX Risk Consulting | EPFL Lausanne

None of these models tell you "where the underlying will be tomorrow". You are confusing Q probability measure (risk neutral, induced from no arbitrage of derivatives prices) and P probability measure (real world probability). Probably the most common mistake in quantitative finance.

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Carlo Acerbi You are right, and that distinction is what I tried to the point in the post. Black-Scholes prices an option under $\mathbb{Q}$ and says nothing about where the underlying will be under $\mathbb{P}$. The Fokker-Planck equation describes the evolution of the density under whichever measure you work in, but the confusion between the two is real and worth making explicit. Thank you for raising it.

Bernard Gourion · 2nd
Head of Research and Development and Quantitative Watch

Well, now a problem Titouan Baraud : do you know that Black scholes is also a model with stochastic discrètes dividends ? Prove it

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Bernard Gourion The standard Black-Scholes framework assumes a continuous, deterministic dividend yield. Extending it to stochastic discrete dividends requires adjusting the forward price and, depending on the model, modifying the underlying SDE, the lognormality assumption no longer holds cleanly across ex-dividend dates. It is a meaningful extension, but one that goes beyond the scope of my post. It's an interesting point that could be extended in a future post tho, and comments are necessary to improve my knowledge!

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